Sofia Domracheva

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MSc student in High Performance Computing with a first-author IEEE paper on quantum vs. classical kernel methods for LHC data and 4+ years of software engineering experience in Python. Interested in hybrid quantum–classical data analysis with ROOT.

Education

- **Sofia University "St. Kliment Ohridski"** — Sofia, Bulgaria
  Master's Degree in High Performance Computing — *Oct. 2026 – Jul. 2028 (expected)*
  - EUMaster4HPC, taught in English. Relevant courses: HPC Architectures, Introduction to Quantum Technologies, Statistics and Machine Learning, Virtualization and Cloud Computing.
- **Plovdiv University "Paisii Hilendarski"** — Plovdiv, Bulgaria
  Bachelor's Degree in Software Technologies and Design; GPA: 5.60/6.00 (Excellent) — *Sep. 2022 – Jul. 2026*

Research Experience

- **Plovdiv University "Paisii Hilendarski"** — Plovdiv, Bulgaria
  Undergraduate Researcher, Quantum Machine Learning (B.Sc. thesis) — *Feb. 2025 – Sep. 2026*
  - **Data Pipeline**: Formulated a binary classification task of jet reconstruction quality on simulated ATLAS Open Data (hadronic jets): data access with ROOT/PyROOT, preprocessing in scikit-learn (standardization, PCA, rescaling for angle encoding).
  - **Quantum Kernels**: Implemented a hybrid quantum-classical SVM (quantum Gram matrix + classical optimization) with a ZZ feature map in Qiskit and benchmarked it against classical RBF and linear SVMs; evaluated the kernel on a GPU-accelerated simulator (Qiskit Aer), a noise model from live `ibm_marrakesh` calibration, and the real device via Qiskit Runtime (demonstration kernel on `ibm_marrakesh` within a 10-minute QPU budget, mean absolute deviation 0.010 vs. ideal).
  - **Results**: The classical RBF SVM reached ROC-AUC 0.88–0.89 vs. at most 0.72 for the QSVM. Measured exponential kernel concentration directly and showed that the gap persists even in noise-free simulation, so better hardware alone would not close it.
  - **Code**: github.com/sofia-domracheva/quantum-jets

Publications

- **S. Domracheva**, A. Penev, S. Stavrev, "Comparative Analysis of Classical and Quantum Support Vector Machines for Large Hadron Collider Data Classification," in *Proc. Int. Conf. Automatics and Informatics (ICAI'26)*, Varna, Bulgaria, Oct. 2026, IEEE (to appear in IEEE Xplore).

Work Experience

- **A Data Pro** — Burgas, Bulgaria
  Software Engineer — *Sep. 2021 – Feb. 2026*
  - **Data Visualization & Backend**: Developed features for a Django/PostgreSQL backend and Python-based visualizations of large datasets.
  - **Infrastructure & Integrations**: Implemented a weekly Celery backup task that archived reports; integrated third-party web services (Google OAuth, reCAPTCHA); worked in a team using internal task-tracking systems.

Projects

- **Geospatial Analysis Platform (NOIT Competition)**: Backend and data pipelines for a retail-location platform: processed OpenStreetMap (Overpass API) and GIS shapefile data to compute regional business density (Python, SQLite).
- **News Aggregator Telegram Bot (European Researchers' Night 2020)**: 2nd place in the "My Chatbot" competition, Burgas. Python, BeautifulSoup, SQLite.

Programming Skills

- **Languages**: Python, SQL, JavaScript
- **Technologies**: ROOT/PyROOT, Qiskit (Aer, Runtime, Machine Learning), NumPy, scikit-learn, Matplotlib, Linux, Git, Django, Celery, PostgreSQL